

Mriganka Ghosh

IoT Automation Developer | Data Analyst | Industrial Control Systems

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[Mriganka Ghosh | LinkedIn](#) [Mriganka Ghosh | Portfolio](#)

PROFESSIONAL EXPERIENCE

Automation Developer / Data Analyst April 2025 – Present
Instadata Pvt. Ltd. | Bengaluru, Karnataka, India

- Designed and deployed Node-RED automation flows for real-time industrial machine monitoring.
- Built VBA macros and Excel add-ins for ABB 800xA batch reporting at pharmaceutical manufacturing sites.
- Developed Python scripts for automated data extraction, transformation, and loading (ETL).
- Integrated IIoT devices using Modbus RTU protocol with alarm logic and multi-device workflows.
- Implemented MQTT-based telemetry pipelines to AWS IoT for remote machine monitoring.
- Managed PostgreSQL and SQLite databases for machine telemetry and batch process data.
- Automated HMI configuration workflows, significantly reducing manual data processing time.
- Debugged and resolved critical production issues in OPC HDA data retrieval and event logging.

KEY PROJECTS

ABB 800xA Batch Reporter

- Developed an Excel-VBA add-in for automated batch data retrieval and trend plotting from ABB 800xA DCS systems.
- Integrated OPC HDA and OPC DA protocols for historical and real-time process tag retrieval.

Node-RED Industrial Machine Monitoring System

- Built multi-device Modbus RTU automation flows with real-time alarm logic and dashboard visualization.
- Integrated PostgreSQL for time-series data storage and published telemetry to AWS IoT Core via MQTT.

MQTT Consumer Service Automated Test Suite

- Developed an automated HTML/JavaScript test suite covering broker management, topic management, and monitoring.
- Produced a comprehensive Postman API testing guide for the MQTT consumer service.

Attendance System using Face Recognition (Academic – Group Leader)

- Led a team to design and build a face-recognition-based attendance application.

TECHNICAL SKILLS

Automation & IIoT

• Node-RED (Flow Development)	• IIoT Device Integration
• Modbus RTU	• MQTT / AWS IoT Core

Programming & Scripting

• Python (ETL, Automation, Scripting)	• JavaScript (Node.js, Web Development)
• VBA / Excel Macros & Add-ins	• SQL (PostgreSQL, SQLite)

Data & Cloud

• MongoDB / Studio3T	• AWS IoT/ AWS S3
• GitHub	• Basic of AI Prompting & Prompt Engineering

Soft Skills

Analytical Problem-Solving	Team Collaboration & Leadership
Technical Communication	Adaptability

CERTIFICATIONS

- AWS S3 Basics - Basic Python for Data Engineering - Introduction to Prompt Engineering for Generative AI - Basic of Generative AI

EDUCATION

Master of Computer Applications (MCA) Brainware University, Kolkata, India CGPA: 7.83	Aug 2022 – Aug 2024
Bachelor of Computer Applications (BCA) Brainware University, Kolkata, India CGPA: 8.63	Aug 2019 – Jul 2022
Higher Secondary (Class XII) – 62.2% Dhuliyān High Madrasah, Murshidabad, West Bengal	Completed 2019
Secondary School (Class X) – 59.1% Kanchantala J.D.J. Institution, Murshidabad, West Bengal	Completed 2017

LANGUAGES

English Bengali Hindi
